

Digital Triage Pseudocode

1. Introduction

This document presents a pseudocode algorithm for digital triage in emergency medical settings. The algorithm is designed to classify patients into one of five Emergency Severity Index (ESI) levels based on vital signs and clinical parameters. The goal is to provide a structured, systematic approach to patient prioritization that can be implemented in digital health systems.

2. Algorithm Overview

The triage algorithm evaluates patient vital signs in a hierarchical manner, starting with the most critical indicators. Each decision point leads to either a triage level assignment or further evaluation. The algorithm considers the following vital signs: systolic blood pressure (SBP), oxygen saturation (SpO2), pulse rate, respiratory rate, temperature, and level of consciousness.

3. Main Triage Algorithm

```
FUNCTION determine_triage_level(patient):  
  // Step 1: Check for immediate life threats  
  IF patient.SBP < 90 OR patient.SpO2 < 90% THEN  
    RETURN ESI_LEVEL_1 // Resuscitation  
  // Step 2: Check for emergent conditions  
  IF patient.SBP < 100 OR patient.SpO2 < 92% THEN  
    RETURN ESI_LEVEL_2 // Emergent  
  // Step 3: Evaluate respiratory distress  
  IF patient.Resp_Rate > 24 OR patient.Resp_Rate < 10 THEN  
    IF patient.SpO2 < 95% THEN  
      RETURN ESI_LEVEL_2  
    ELSE  
      RETURN ESI_LEVEL_3 // Urgent  
    END IF  
  END IF  
  // Step 4: Calculate vital score  
  vital_score = calculate_vital_score(patient)  
  IF vital_score >= HIGH_RISK THEN  
    RETURN ESI_LEVEL_3  
  ELSE IF vital_score >= MODERATE_RISK THEN
```

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RETURN ESI_LEVEL_4 // Less Urgent
ELSE
RETURN ESI_LEVEL_5 // Non-Urgent
END IF
END FUNCTION

```

4. Vital Score Calculation

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FUNCTION calculate_vital_score(patient):
score = 0
// Pulse rate scoring
IF patient.Pulse > 120 OR patient.Pulse < 60 THEN
score = score + 3
ELSE IF patient.Pulse > 100 OR patient.Pulse < 70 THEN
score = score + 1
END IF
// Temperature scoring
IF patient.Temp > 39.0 OR patient.Temp < 35.0 THEN
score = score + 3
ELSE IF patient.Temp > 38.0 OR patient.Temp < 36.0 THEN
score = score + 1
END IF
// Respiratory rate scoring
IF patient.Resp_Rate > 24 OR patient.Resp_Rate < 10 THEN
score = score + 3
END IF
RETURN score
END FUNCTION

```

5. ESI Level Definitions

Level 1	Resuscitation	Immediate life-threatening condition	Immediate
Level 2	Emergent	High-risk; severe pain/distress	< 10 min
Level 3	Urgent	Stable but needs multiple resources	< 30 min
Level 4	Less Urgent	Stable; needs one resource	< 60 min

Level 5	Non-Urgent	Stable; needs no resources	< 120 min
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6. Vital Signs Reference Ranges

Pulse Rate	60-100 bpm	100-120 or 50-60	>120 or <50
Resp Rate	12-20 /min	20-24 or 10-12	>24 or <10
Temperature	36.0-38.0 C	38.0-39.0 or <36.0	>39.0 or <35.0
Systolic BP	110-140 mmHg	140-180 or 90-110	>180 or <90
SpO2	95-100%	92-95%	<92%

7. Implementation Notes

This pseudocode provides a framework for implementing digital triage in clinical decision support systems. The algorithm should be used in conjunction with clinical judgment and may require adjustment based on local protocols and patient populations.